Code No: 133AR/123AU/113AU/53009 **R16/R15/R13/R09**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year I Semester Examinations, March - 2021
ELECTRONIC DEVICES AND CIRCUITS
(R16 - Common to EIE, MCT; R15 - Common to EEE, ECE, CSE, IT, MCT;
R13 - Common to EEE, ECE, CSE, EIE, IT, MCT; R09 - Common to EEE, ECE, CSE, IT)

Time: 3 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks

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1.a) Draw the V-I characteristics of diode and explain its temperature dependency.
b) Neglecting the effect of carrier generation and recombination in the space-charge region, find an expression for the diffusion capacitance of an ideal p-n junction diode in which the doping concentrations of the p and n sides are comparable to each other. [7+8]

2.a) Explain the principle of operation of varactor diode and also write its applications.
b) How Zener diode is used for voltage regulation? Draw the circuit diagram. [8+7]

3.a) Derive an expression for the ripple in a ∏-section filter when used with a halfwave rectifier.
b) The output of a full wave rectifier is fed from a 40-0-40V transformer. The load current is 0.1A. Two 40μF capacitances are available. The circuit resistance exclusive of the load is 50Ω. Calculate value of inductance for a two stage L-section filter. The inductances are to be equal. The ripple factor is to be 0.0001. [8+7]

4.a) What are the different transistor current components for a forward biased emitter junction and a reverse biased collector junction and derive the equations for emitter efficiency, transport factor and large signal current gain.
b) Draw the common base configuration circuit, input characteristics, output characteristics and identify different regions of operation. [8+7]

5.a) Show that the exact expression for h_{fe} in terms of the CB hybrid parameters is

$$h_{fe} = -\frac{h_{fb}(1-h_{rb}) + h_{ib}h_{ob}}{(1+h_{fe})(1-h_{rs}) + h_{ob}h_{ib}}$$

b) Given a single stage transistor amplifier with the h parameters h_{ic} =1.1K, h_{fc}= -51, h_{rc}=1, h_{oc}= 25μA/V, calculate A_I, A_V, A_{vs}, R_i and R_o for the CC transistor configuration, with $R_s = R_L$ = 10K. [7+8]

6.a) Determine the stability factor for the fixed bias circuit and compare its merits and demerits with other biasing circuits.
b) A transistor with β=100 is to be used in the collector-emitter feedback bias configuration with R_c=1.5K, R_e=2K,R_b=370K, V_{CC}=12V,V_{BE}=0.6V and I_{C0}=0. Calculate the collector to emitter voltage V_{CE} and also find the stability factor? [7+8]

7.a) How a thermistor and sensistor are used for transistor bias compensation?
b) What is meant by thermal instability in a transistor? How to achieve stability? [10+5]

8.a) Draw the V-I characteristics of enhancement MOSFET and explain it.
b) Consider the common drain amplifier and assume the FET is properly biased and the FET has the following parameters: g_m=2mA/V and r_d=100K. Using the small signal low frequency model derive an expression for the gain of the amplifier. [7+8]

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